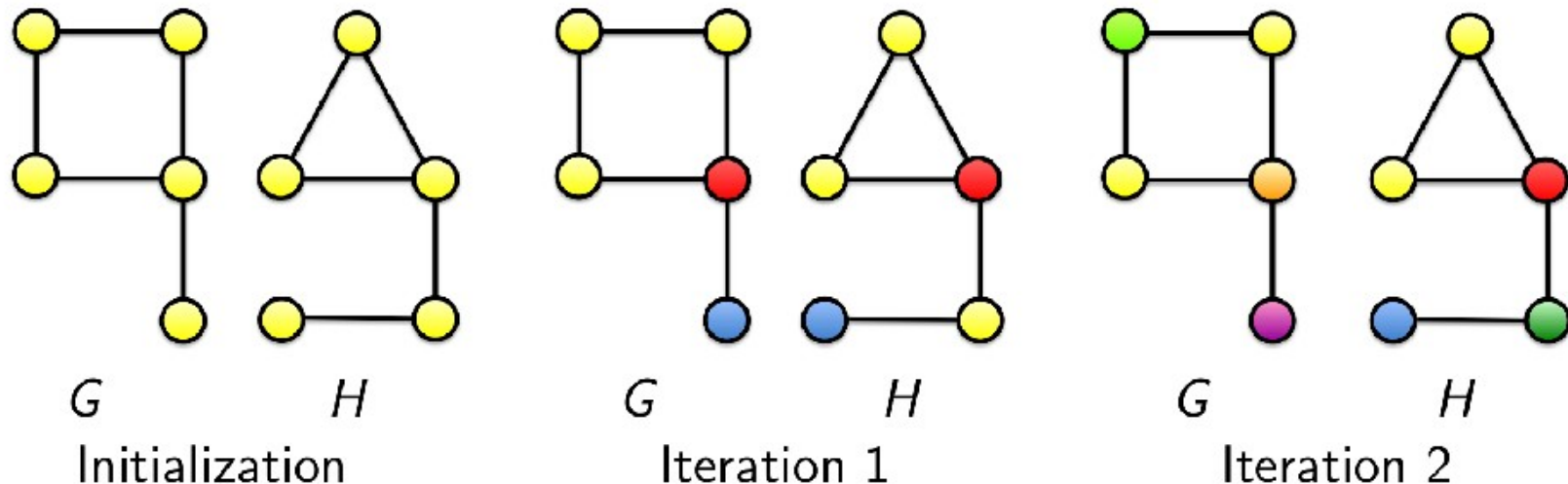


Graph Isomorphism

- Not known to be NP-hard, but no polynomial algorithm known either
- Existing exact algorithms are hence **search algorithms** that search through the space of possible mappings between nodes
- To reduce the number of possible mappings, they use a form of propagation known as **color refinement** (or the Weisfeiler-Leman algorithm)

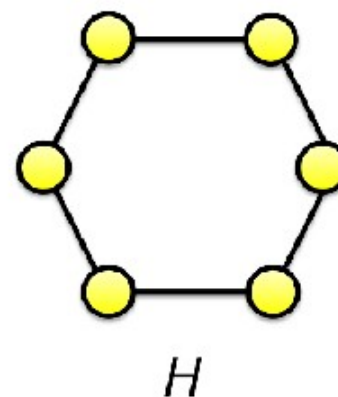
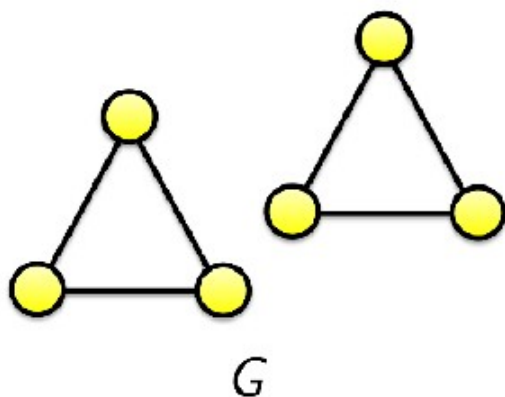
Color Refinement

- **Initial:** all vertexes have the same color
- **Iteration:** Further separation of colors based on color histograms of neighbors (“value of node updated based on messages from the neighbors”)
- Stop when no nodes with the same color can be separated further

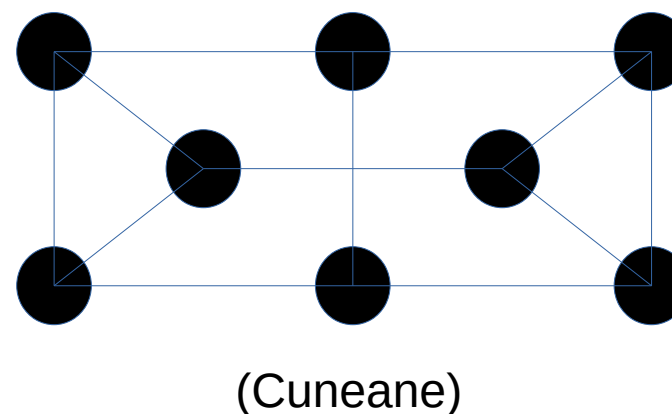
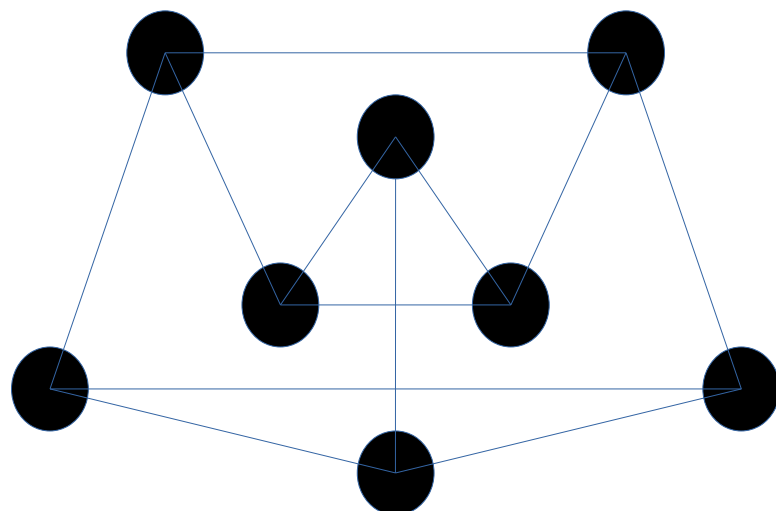


WL for Graph Isomorphism

- WL cannot distinguish these two *regular* graphs:



- Or between these two molecules:

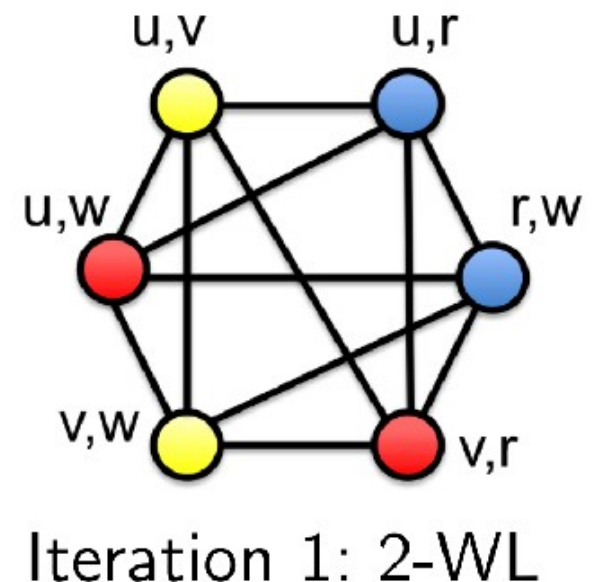
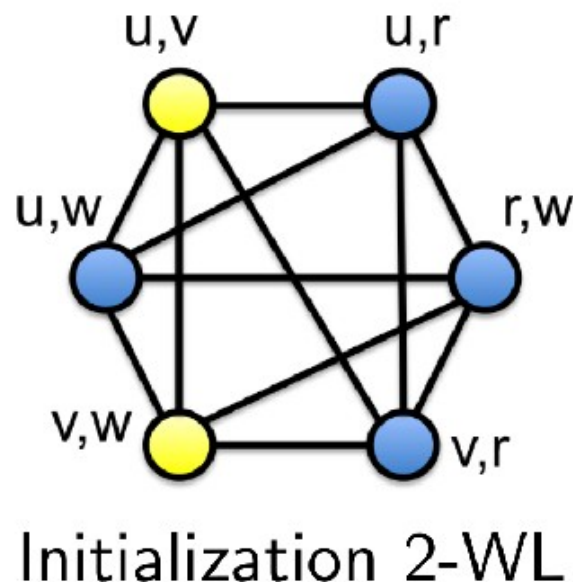
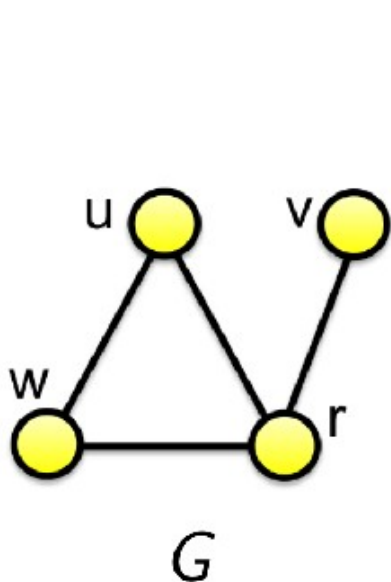


WL for Graph Isomorphism

- If two graph are isomorphic $\rightarrow$ WL will not find difference between the graphs
- If WL does not find a difference between graphs $\rightarrow$ it is still possible that the two graphs are not isomorphic
- Hence, additional search remains necessary to decide that two graphs are isomorphic

k-WL for Graph Isomorphism

- Two *k*-sets are neighbors if they differ in one element
- **Initial:** *k*-sets *U*, *W* get the same color if $G[U] \simeq G[W]$.
- **Iteration:** Two identically colored *k*-sets *U* and *W* get different colors if there exists a color *c* so that *U* and *W* have a different cardinality set of neighbors of color *c*.



Power of k -WL

- For $k > 2$ k -WL is stronger than WL
- Exact for large k (for instance if $k = n$, by definition)
- For $k = 3$ regular graphs can be distinguished

